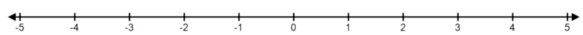
Problem 6

Problem 6

Problem 1 *Consider an object moving along a line with velocity* $v(t) = \pi \sin(\pi t)$ *on* $[0, 2]$ *and initial position* $s(0) = 0$. *Time is measured in seconds and velocity in m/s.*

(a) *Determine the position function,* $s(t)$, *on* $[0, 2]$.

(b) *Mark the position of the object at the time* $t = 1$ *on the line below.*



(c) *Determine the average velocity,* v_{av}, *of the object during the interval* $[0, 2]$.

(d) *Determine when the motion is in the positive direction.*

(e) *At what time (or times) is the object farthest from the origin?*